JAVA ENTERPRISE PRACTICALS

Practical 7: JSP Implicit Objects

Environment: Apache NetBeans · GlassFish Server · JDK 21

1. Aim

To demonstrate the use of JSP implicit objects — `request`, `response`, `out`, and `session` — by displaying request information, response headers, output methods, and session attributes.

2. Theory

JSP provides several predefined objects that are automatically available in scriptlets and expressions. These are called **implicit objects**. They give access to the request, response, output stream, session, and other container-managed resources without requiring explicit declaration.

This practical demonstrates four important implicit objects:

- **request** — Represents the HTTP request (method, server name, port, parameters).
- **response** — Represents the HTTP response (content type, encoding, headers).
- **out** — The `JspWriter` used to send output to the client (`println`, `print`, `write`).
- **session** — Represents the user's HTTP session for storing attributes across requests.

3. Complete Steps

1. Open Apache NetBeans IDE.
2. Select **File** → **New Project**.
3. Choose **Java with Ant** → **Web** → **Web Application** (or the equivalent Web Application template in your version).
4. Enter the project name **JSPImplicitObjects**.
5. Select GlassFish Server and a compatible Java EE/Jakarta EE version.
6. Click **Finish**.
7. Expand the project and open the **Web Pages** folder. In a Maven project, use `src/main/webapp`.
8. Create a JSP file named `implicitDemo.jsp`.
9. Enter the complete code given below.
10. Save the file.
11. Right-click the project and select **Run**.
12. If the browser does not open automatically, visit:
`http://localhost:8080/JSPImplicitObjects/implicitDemo.jsp`
13. Verify the output for all four implicit objects: `request`, `response`, `out`, and `session`.

4. Complete Code: implicitDemo.jsp

Create this JSP file under Web Pages (or `src/main/webapp`).

```
<%@page language="java"
        contentType="text/html"
        pageEncoding="UTF-8" %>

<!DOCTYPE html>
<html>
<head>
    <title>JSP Implicit Objects</title>
</head>
<body>
```

```
    <h1>Demonstration of JSP Implicit Objects</h1>

    <hr>
    <h2>1. Request Object</h2>

    <%
        out.println("Method: " + request.getMethod() + "<br>");
        out.println("Server Name: " + request.getServerName() + "<br>");
        out.println("Server Port: " + request.getServerPort() + "<br>");
    %>

    <hr>
    <h2>2. Response Object</h2>

    <%
        out.println("Content Type: " + response.getContentType() + "<br>");
        out.println("Character Encoding: "
                    + response.getCharacterEncoding() + "<br>");

        response.setHeader("X-Practical", "JSP-Implicit-Objects");

        out.println("Response header set successfully.<br>");
    %>

    <hr>
    <h2>3. Out Object</h2>

    <%
        out.println("This message is displayed using out.println().<br>");
        out.print("This message is displayed using out.print().<br>");
        out.write("This message is displayed using out.write().<br>");
    %>

    <hr>
    <h2>4. Session Object</h2>

    <%
        session.setAttribute("studentName", "Rahul");
        session.setAttribute("course", "Java Enterprise");

        out.println("Session ID: " + session.getId() + "<br>");
        out.println("Student Name: "
                    + session.getAttribute("studentName") + "<br>");
        out.println("Course: "
                    + session.getAttribute("course") + "<br>");
    %>

</body>
</html>
```

5. Important Code Explanation

Object / Method	Purpose
request.getMethod()	Returns the HTTP method (GET, POST, etc.).
request.getServerName()	Returns the server host name.
request.getServerPort()	Returns the server port number.
response.getContentType()	Returns the content type of the response.
response.getCharacterEncoding()	Returns the character encoding (e.g. UTF-8).
response.setHeader(...)	Sets a custom response header.
out.println() / out.print()	Writes text (with or without a newline) to the client.

Object / Method	Purpose
out.write()	Writes a string to the output stream.
session.setAttribute(...)	Stores a value in the user's session.
session.getAttribute(...)	Retrieves a value previously stored in the session.
session.getId()	Returns the unique session identifier.

6. Expected Output

When you open the JSP page, the browser displays sections similar to the following (exact values depend on your server configuration):

```
Demonstration of JSP Implicit Objects

1. Request Object
Method: GET
Server Name: localhost
Server Port: 8080

2. Response Object
Content Type: text/html;charset=UTF-8
Character Encoding: UTF-8
Response header set successfully.

3. Out Object
This message is displayed using out.println().
This message is displayed using out.print().
This message is displayed using out.write().

4. Session Object
Session ID: (a unique session identifier)
Student Name: Rahul
Course: Java Enterprise
```

7. Key Points to Remember

1. JSP implicit objects are created automatically by the container; you do not need to declare them.
2. `request` provides information about the current HTTP request.
3. `response` is used to control headers, content type, and status.
4. `out` is a `JspWriter` for writing HTML/text to the client.
5. `session` stores user-specific data across multiple requests.
6. Session attributes persist until the session times out or is invalidated.

8. Result

A single JSP page was created that demonstrates the four core implicit objects: request, response, out, and session. The page displays request details, response properties, different output methods, and session attributes, confirming that implicit objects work as expected in the GlassFish environment.